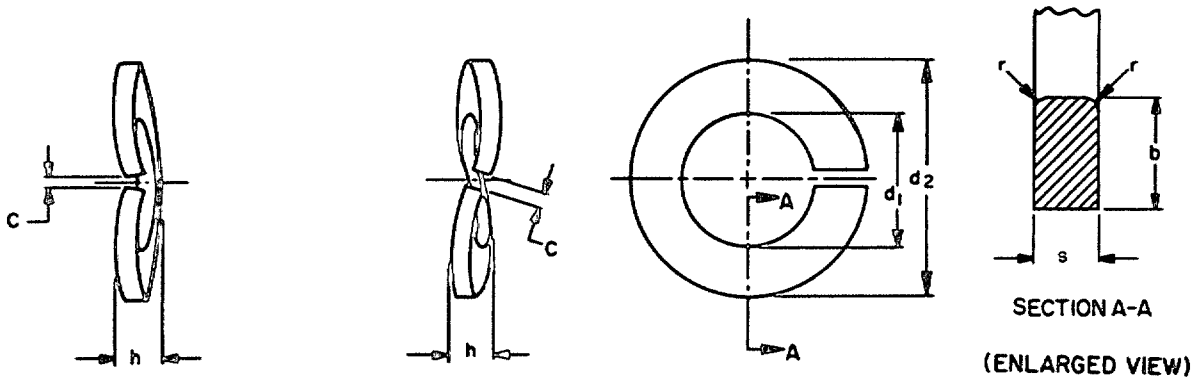
MILITARY SPECIFICATION SHEET

WASHER, SPRING TENSION, CURVED OR CRINKLE TYPE, (HIGH COMPRESSION), METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the washers described herein shall consist of this document and the latest issue of DOD-W-63474.



WASHER, CURVED
SEE TABLE I

WASHER, CRINKLE (WAVED)
SEE TABLE I

NOTE:
1. All dimensions are in millimeters,
2. Break all edges, remove all burrs.

FIGURE 1. WASHER, SPRING TENSION, CURVED OR CRINKLE TYPE, (HIGH COMPRESSION) METRIC

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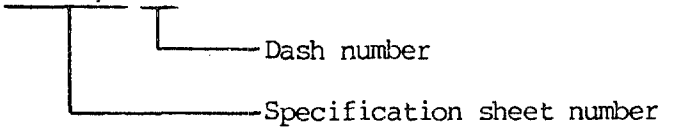
REQUIREMENTS:

1. Material: Spring steel, Grades 1070 thru 1095 (UNS G10700 thru UNS G10950).

2. Protective coating and surface treatment: Cadmium plate per QQ-P-416, Type II, Class 3.

3. Part number: The part number consists of the letter "D" the basic number of the specification sheet, followed by a dash number taken from Table I.

Example: D63474/4-01



D63474/4-01 indicates: Washer, Spring Tension, Curved (High Compression) Cadmium Plated; 2.1 mm ID; 4.4 mm OD; 0.5 mm Thick.

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TABLE I. DASH NUMBERS AND DIMENSIONS

FOR THREAD DIAMETER	DASH NUMBERS CURVED	DASH NUMBERS CRINKLE (WAVED)	d_1	d_1 TOL	d_2 max	h min	h max	b	b TOL	s	s TOL	r ≈
2	01	31	2.1		4.4	0.7	0.9	0.9		0.5		
2.3	02	32	2.4	+0.3	4.9	0.9	1.1	1	±0.1	0.6	±0.1	0.1
2.5	03	33	2.6		5.1	0.9	1.1	1		0.6		
2.6	04	34	2.7		5.2	0.9	1.1	1		0.6		
3	05	35	3.1	+0.3	6.2	1.1	1.3	1.3	±0.1	0.7	±0.1	0.1
3.5	06	36	3.6		6.7	1.1	1.3	1.3		0.7		
4	07	37	4.1	+0.3	7.6	1.2	1.4	1.5	±0.1	0.8	±0.1	0.2
5	09	38	5.1		9.2	1.5	1.7	1.8		1		
6	09	39	6.1	+0.4	11.8	2.0	2.2	2.5	±0.15	1.3		0.3
7	10	40	7.1	+0.4	12.8	2.0	2.2	2.5	±0.15	1.3	±0.1	0.3
8	11	41	8.2		14.9	2.45	2.75	3		1.6		0.5
10	12	42	10.2	+0.5	18.1	2.85	3.15	3.5	±0.2	1.8		
12	13	43	12.2	+0.5	21.1	3.35	3.65	4		2.1		
14	14	44	14.2		24.1	3.9	4.3	4.5	±0.2	2.4	±0.15	1
16	15	45	16.2	+0.8	27.4	4.5	5.1	5		2.8		
18	16	46	18.2	+0.8	29.4	4.5	5.1	5		2.8	±0.15	
20	17	47	20.2	+1	33.6	5.1	5.9	6	±0.2	3.2	±0.2	1
22	18	48	22.5		35.9	5.1	5.9	6		3.2		
24	19	49	24.5	+1	40	6.5	7.5	7		4		
27	20	50	27.5		43	6.5	7.5	7	±0.25	4	±0.2	1.6
30	21	51	30.5	+1.2	48.2	9.5	10.5	8		6		
36	22	52	36.5	+1.2	58.2	10.3	11.3	10		6	±0.2	1.6
42	23	53	42.5		68.2	12	13	12	±0.25	7	±0.25	2
48	24	54	49	+1.5	75	12	13	12		7		
56	25	55	57	+1.5	87	14	15.5	14		8		
64	26	56	65		95	14	15.5	14	±0.25	8	±0.25	2
72	27	57	73		103	14	15.5	14		8		
80	28	58	81	+1.5	111	14	15.5	14		8		
90	29	59	91		121	14	15.5	14	±0.25	8	±0.25	2
100	30	60	101		131	14	15.5	14		8		

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TABLE II. TEST LOADS FOR CURVED AND CRINKLE WASHERS

SIZE	2	2.3	2.5	2.6	3	3.5	4	5	6	7	8	10
LOAD N	10.0	12.5	12.5	12.5	15.0	22.0	30.0	50.0	70.0	100.0	120.0	220.0
HEIGHT OF SPRING WASHER AFTER REMOVAL OF LOAN MIN	0.89	1.0	1.0	1.0	0.85	0.85	0.95	1.18	1.5	1.5	2	2.5

SIZE	12	14	16	18	20	22	24	27	30	36	42 AND 48	56 TO 100
LOAD N	320.0	450.0	650.0	760.0	1000.0	1300.0	1450.0	1950.0	2400.0	3500.0	3500.0	3500.0
HEIGHT OF SPRING WASHER AFTER REMOVAL OF LOAD MIN	2.65	3	3.6	3.6	4.2	4.2	5.3	5.3	7.6	7.6	9	10.5

NOTE: Loading and prolonged loading tests shall be conducted in accordance with DoD-W-63474, paragraphs 4.4.3.5 Loading and 4.4.3.6 Prolonged Loading by application of the loads provided by Table II above. Loading time for test prescribed by paragraph 4.4.3.5 Loading, shall be three minutes.

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AT, AV
Navy - MC
Air Force - 18
NASA - JFK
NSA - NS
DLA - IS

User activities:
Navy - YD

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5310-1474)

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